Problem Set 31

Evaluating Integrals

1 Evaluate the following definite and indefinite integrals; please show your work. It will sometimes be helpful to simplify your functions prior to looking for an antiderivative.

Although FTOC 2 is very useful for evaluating definite integrals, we have learned other ways of evaluating definite integrals; those methods are still useful for some of the integrals below! Remember that you can always check your indefinite integrals by differentiating them.

(a) $\displaystyle\int \left(\frac{1}{2x} + \frac{e}{\pi x^2}\right) dx$

Solution

We can rewrite this as

$$\begin{aligned}\int \left(\frac{1}{2x} + \frac{e}{\pi x^2}\right) dx &= \frac{1}{2}\int \frac{1}{x}\,dx + \frac{e}{\pi}\int x^{-2}\,dx \\ &= \frac{1}{2}\ln|x| + \frac{e}{\pi}\left(-x^{-1}\right) + C \\ &= \boxed{\frac{1}{2}\ln|x| - \frac{e}{\pi x} + C}\end{aligned}$$

(b) $\displaystyle\int_{-\pi/6}^{\pi/4} \left(\sec(\theta)\tan(\theta) + \frac{\sec^2(\theta)}{\sqrt{3}}\right) d\theta$

Solution

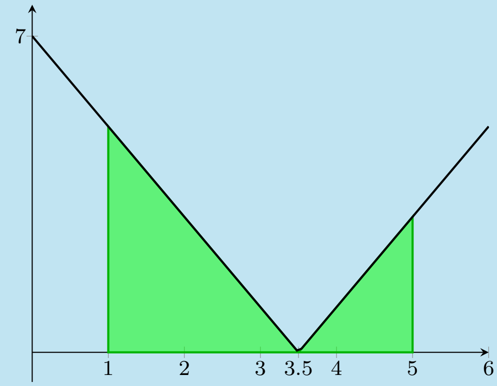
It helps to remember that $\frac{d}{d\theta}\sec\theta = \sec(\theta)\tan(\theta)$, or at least know where to look it up.

$$\begin{aligned}\int_{-\pi/6}^{\pi/4} \left(\sec(\theta)\tan(\theta) + \frac{\sec^2(\theta)}{\sqrt{3}}\right) d\theta &= \sec\theta + \frac{1}{\sqrt{3}}\tan\theta\Big|_{-\pi/6}^{\pi/4} \\ &= \left(\sqrt{2} + \frac{1}{\sqrt{3}}\right) - \left(\frac{2}{\sqrt{3}} - \frac{1}{3}\right) \\ &= \boxed{\sqrt{2} - \frac{1}{\sqrt{3}} + \frac{1}{3}}\end{aligned}$$

(c) $\displaystyle\int_1^5 |2t-7|\,dt$

Solution

We saw in Problem Set 30 that antiderivatives of absolute value functions are complicated. Let's instead try to visualize this graphically; this is asking for the signed area under $y = |2t-7|$ from $t = 1$ to $t = 5$:



The left triangle has base $\frac{5}{2}$ and height 5 while the right triangle has base $\frac{3}{2}$ and height 3. So, the area is $\frac{25}{4} + \frac{9}{4} = \boxed{\frac{17}{2}}$.

(d) $\displaystyle\int \frac{u^2 + \ln \pi}{\sqrt[4]{u}}\, du$

Solution

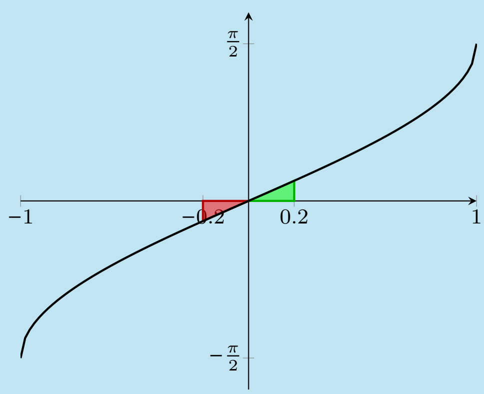
We can rewrite this as

$$\begin{aligned}\int \frac{u^2 + \ln \pi}{\sqrt[4]{u}}\, du &= \int u^{7/4}\, du + (\ln \pi) \int u^{-1/4}\, du \\ &= \boxed{\frac{4}{11} u^{11/4} + \frac{4}{3} (\ln \pi) u^{3/4} + C}\end{aligned}$$

(e) $\displaystyle\int_{-0.2}^{0.2} \arcsin x\, dx$

Solution

We don't know an antiderivative of $\arcsin x$, so we'll try a graphical approach. Since $\arcsin x$ is an odd function, this signed area is $\boxed{0}$ by symmetry:



(f) $\displaystyle\int_0^{-2} \left(\sqrt{4 - x^2} + 3e^x\right) dx$

Solution

$\int \sqrt{4-x^2}\,dx$ is not one of the basic indefinite integrals that we saw on the worksheet from class; however, you should recognize $\sqrt{4-x^2}$ as a function whose graph is simple geometrically: it's a semicircle. So, let's split the integral up and use two different strategies:

$$\int_0^{-2} \left(\sqrt{4-x^2} + 3e^x\right) dx = \int_0^{-2} \sqrt{4-x^2}\,dx + 3\int_0^{-2} e^x\,dx.$$

We can visualize $\int_0^{-2} \sqrt{4-x^2}\,dx$ as $-\int_{-2}^{0} \sqrt{4-x^2}\,dx$, and $\int_{-2}^{0} \sqrt{4-x^2}\,dx$ is the area of a quarter circle of radius 2, so it's equal to $\frac{1}{4}\cdot\pi\cdot 2^2 = \pi$. On the other hand, $\int_0^{-2} e^x\,dx = e^x\Big|_0^{-2} = e^{-2} - 1$. Putting this together, $\int_0^{-2}\left(\sqrt{4-x^2} + 3e^x\right) dx = \boxed{-\pi + 3e^{-2} - 3}$.

2 (a) Find antiderivatives for the given functions. In other words, for each function f, find a function F such that $F' = f$.

i. $f(x) = e^{3x}$

Solution

We can take $F(x) = \boxed{\frac{1}{3}e^{3x}}$ (plus any constant).

ii. $f(x) = \dfrac{3}{e^x}$

Solution

We can rewrite $f(x) = 3e^{-x}$, so $F(x) = \boxed{-3e^{-x}}$ works.

iii. $f(x) = \left(\sqrt[3]{x^2} + 7x^2\right)\left(\pi\sqrt{x^5} + e\right)$

Solution

Since we don't have a product rule for antiderivatives, we should rewrite f; we can expand it as

$$f(x) = \left(x^{2/3} + 7x^2\right)\left(\pi x^{5/2} + e\right) = \pi x^{19/6} + 7\pi x^{9/2} + ex^{2/3} + 7ex^2,$$

so one antiderivative is $F(x) = \boxed{\frac{6}{25}\pi x^{25/6} + \frac{14}{11}\pi x^{11/2} + \frac{3}{5}ex^{5/3} + \frac{7ex^3}{3}}$.

iv. $f(x) = \dfrac{(3e^x)^2}{\sqrt{5^x}}$

Solution

We don't have a Quotient Rule for antiderivatives, so let's first rewrite this: $f(x) = \dfrac{9e^{2x}}{(\sqrt{5})^x} = 9\left(\dfrac{e^2}{\sqrt{5}}\right)^x$, so one antiderivative is $F(x) = \boxed{\dfrac{9}{\ln(e^2/\sqrt{5})}\left(\dfrac{e^2}{\sqrt{5}}\right)^x = \dfrac{9}{2-\frac{1}{2}\ln 5}\left(\dfrac{e^2}{\sqrt{5}}\right)^x}$.

(b) Check your answers to (a) by differentiating them.

Solution

i. The derivative of $F(x) = \frac{1}{3}e^{3x}$ is $\frac{1}{3}\cdot 3e^{3x} = e^{3x}$.

ii. The derivative of $F(x) = -3e^{-x}$ is $-3\left(-e^{-x}\right) = 3e^{-x}$.

iii. The derivative of $F(x) = \dfrac{6}{25}\pi x^{25/6} + \dfrac{14}{11}\pi x^{11/2} + \dfrac{3}{5}ex^{5/3} + \dfrac{7ex^3}{3}$ is $\pi x^{19/6} + 7\pi x^{9/2} + ex^{2/3} + 7ex^2$.

iv. The derivative of $F(x) = \dfrac{9}{\ln(e^2/\sqrt{5})}\left(\dfrac{e^2}{\sqrt{5}}\right)^x$ is $9\left(\frac{e^2}{\sqrt{5}}\right)^x$.

3 The Long Trail in Vermont is the oldest long-distance trail in the United States. Suppose we measure positions on the trail relative to Camel's Hump, a peak on the trail. That is, we'll describe positions along the trail by how many miles north of Camel's Hump they are. (So, a location 10 miles south of Camel's Hump would be at position -10.)

One day, Ali sets out from Camel's Hump at noon for a hike. His velocity t hours after noon is

$$v(t) = \frac{3t^2 + 2t - 56}{24} \text{ miles per hour.}$$

(a) Find Ali's net change in position between noon and 7 pm. Use a graph to justify your work.

Solution

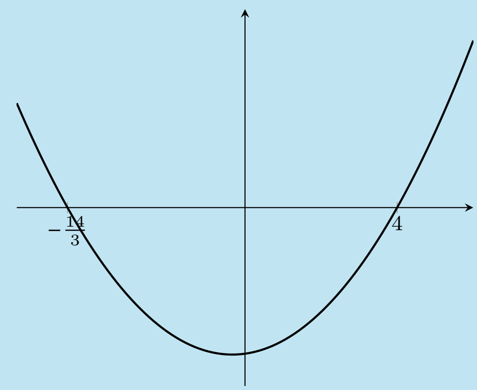
Ali's net change in position between noon and 7 pm is

$$\int_0^7 v(t)\,dt = \left.\frac{t^3 + t^2 - 56t}{24}\right|_0^7 = \boxed{0 \text{ miles}}.$$

(b) Find the total distance Ali hiked between noon and 7 pm.

Solution

Since the rate of change of distance traveled is speed and Ali's speed at time t is $|v(t)|$, the distance Ali hiked between noon and 7 pm is $\int_0^7 |v(t)|\,dt$. Let's think about the graph of $v(t)$; first, $v(t)$ is a quadratic which factors as $\dfrac{1}{24}(3t^2 + 2t - 56) = \dfrac{1}{24}(3t+14)(t-4)$, so it has roots at $t = 4$ and $t = -\frac{14}{3}$. Therefore, its graph looks like this:



In particular, on $(0,4)$, $v(t)$ is negative, so $|v(t)| = -v(t) = -\dfrac{3t^2+2t-56}{24}$; on $(4,7)$, $v(t)$ is positive, so $|v(t)| = v(t) = \dfrac{3t^2+2t-56}{24}$. So, the total distance Ali travels between $t = 0$ and $t = 7$ is

$$\int_0^4 -\frac{3t^2+2t-56}{24}\,dt + \int_4^7 \frac{3t^2+2t-56}{24}\,dt = \left(-\frac{t^3+t^2-56t}{24}\bigg|_0^4 \right) + \left(\frac{t^3+t^2-56t}{24}\bigg|_4^7 \right)$$
$$= \boxed{12 \text{ miles}}$$

(c) Describe Ali's hike in words. (We're looking for an answer like, "He walked 1 mile north, then turned around at 1 pm and walked 2 miles south, then …".)

Solution

In the first 4 hours of his hike, Ali walked 6 miles south. At 4 pm, he turned around and walked 6 miles north, returning to Camel's Hump at 7 pm.

4 Suppose that a car driver brakes when the car is traveling at 60 mph (which is 88 ft/s). Based on data from the National Highway Traffic Safety Administration, for a typical car, the braking gives the car a constant acceleration of -22 ft/s^2. How long will it take for such a car to come to a complete stop? How far will the car travel before stopping?

Solution

Let's first try to write down the car's velocity v as a function of time t. Since the car has a constant acceleration of -22 ft/s^2, $v'(t) = -22$, so $v(t) = -22t + C$. We can find C by using the fact that the car's velocity right when it starts braking is 88. If we call the time when it starts braking $t = 0$, then $88 = v(0) = -22(0) + C$, so $C = 88$. Therefore, $v(t) = -22t + 88$.

The car comes to a complete stop when $v(t) = 0$, which happens at $t = 4$. That is, it takes

$$\boxed{\text{4 seconds for the car to come to a complete stop}}.$$

The distance the car travels in this time is

$$\int_0^4 v(t)\,dt = 88t - 11t^2 \bigg|_0^4 = \boxed{176 \text{ feet}}.$$

5 **Reflect Back.** When evaluating integrals, one of the most important skills is to recognize antiderivatives that you can do easily. Evaluate **five** of the integrals below. (The rest are difficult! So, your job is to figure out which are easy and to do those.)

(a) $\int \frac{1}{u^2}\,du$

(b) $\int \frac{1}{u^2-1}\,du$

(c) $\int \frac{1}{u^2+1}\,du$

(d) $\int \frac{1}{e^u}\,du$

(e) $\int u^{-1}\,du$

(f) $\int \ln u\,du$

(g) $\int \tan u\,du$

(h) $\int \sin u\,du$

(i) $\int \sin(u^2)\,du$

Solution

The easy ones are:

- $\int \frac{1}{u^2}\,du = \int u^{-2}\,du = -u^{-1} + C$
- $\int \frac{1}{u^2+1}\,du = \arctan u + C$
- $\int \frac{1}{e^u}\,du = \int e^{-u}\,du = -e^{-u} + C$
- $\int u^{-1}\,du = \int \frac{1}{u}\,du = \ln|u| + C$
- $\int \sin u\,du = -\cos u + C$

The other integrals are challenging, but doable with other integration techniques. Except for (i), which actually does not have a nice algebraic formula at all, no matter how good you are at calculus.